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뇌-마음-행동 개별연구 보고서

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연구 제목	Hierarchical Emergence of Emotion in Deep Visual Representations

1. 연구의 배경 및 중요성

Understanding how emotional information is perceived and encoded from visual stimuli remains a fundamental challenge. While modern emotion recognition systems achieve high classification accuracy, the internal mechanisms by which visual systems derive affective cues—especially when emotional signals are weak or subtle—are still poorly understood. This gap is particularly evident when emotional expressions are not consciously detectable by human observers.

In the biological visual system, visual features are hierarchically abstracted along the ventral stream into semantic object representations, which are subsequently interpreted by affective circuits under contextual modulation. Convolutional Neural Networks (CNNs partially mirror this hierarchical organization, especially in early-to-intermediate visual processing stages. However, how emotion-relevant representations emerge, concentrate, and propagate within such hierarchies remains unclear.

This study addresses this gap by analyzing emotion representations across depth in CNN architectures with distinct computational motifs. Specifically, we compare strictly feedforward VGG networks and residual-based ResNet models to examine where emotion-related information arises, how it is structurally organized within layers, and how it is transformed across the network. By systematically characterizing these processes, this work aims to provide interpretable insights into both artificial emotion recognition mechanisms and their correspondence to biological visual-emotional processing.

2. 연구 방법 및 결과

This work is divided into 5 stages. The first one being Stage0 for preliminary analysis using the Resnet18 and VGG16. The purpose of it being establishing the ground truth for which layers contain emotion-relevant information. As explained in the table below, both models extract information from pictures classified into 4 emotional categories: Fear, Happy, Sad and Neutral. Using 5 convolutional layers with its input being 120 images across 3 channels and 224*224 pixels. Using a model using 5 hierarchical layers was very important for this research. It is commonly assumed visual systems take notice of an object representation starting from low features to higher features. Compared to visual systems within the biological brain, comprised of V1 with smaller receptive fields for edge detection, passing the V2, V3 representing higher level features of objects, to finally reaching the V4 area mainly concerning color and texture extraction this information comprises into the

Inferior temporal cortex as well as the amygdala to final object and emotional extraction. CNN(Convolutional Neural Network) models convey similar formats as extracting small local patterns as edges, orientations in earlier layers leading up to texture, local patterns and entire areas in following ones.

This task was able with the introduction of transfer learning of VGG16, and the ResNet18 to the image dataset. Transfer learning can be effectively administered when the dataset is relatively smaller in size compared to the model itself therefore at risk of overfitting. Freezed pre-trained weights make the training on smaller sizes of dataset efficient. To improve computational efficiency, and since emotion recognition and classification tasks share both low-level and high-level features similar to those in visual CNN applications, this approach is beneficial. Approaches in transfer learning include fine-tuning: which unfreezes some layers to free up parts of entire networks to be refitted to the dataset, full-training: not using pre-trained weights rather randomly initialized ones to optimize performance to the given task, and simple feature extraction: fixing all pre-trained weights of the backbone model only adjusting to the new classifier of the novel task. As the task in the research neither requires a large dataset nor a optimization to an entirely new task, uses the simplest means of transfer learning to efficiently check and analyze the result.

(1) VGG16

Layer	Output shape	Spatial res	연산	Channels C
L1	N * 64 * 224 * 224	224 * 224	Block1 ReLu	64
L2	N * 128 * 56 * 56	56 * 56	Block2 Max pool	128
L3	N * 256 * 28 * 28	28 * 28	Block3 Max Pool	256
L4	N * 512 * 28 * 28	28 * 28	Block1 RELu	512
L5	N * 512 * 14 * 14	14 * 14	Block1 pool	512

(2) ResNet18

Layer	Output shape	Spatial res	연산	Channels C
L1	N * 64 * 56 * 56	56 * 56	Conv + bn + relu +max pool	64
L2	N * 64 * 56 * 56	56 * 56	Conv + bn + relu +max pool	64
L3	N * 128 * 28 * 14	28 * 28	Conv + bn + relu +max pool	128
L4	N * 256 * 14 * 14	14 * 14	Conv + bn + relu +max pool	256
L5	N * 512 * 7 * 7	7 * 7	Conv + bn + relu +max pool	512

1) REPRESENTATIONAL INFORMATION ANALYSIS (UI-Based)

Activation tensors were extracted from all five analysis layers of each model and flattened into feature vectors, which were subsequently z-scored. Cosine similarity was computed for every image pair, producing an $N \times N$similarity matrix representing how strongly each layer clusters images of the same emotion. A Representational Dissimilarity Matrix (RDM) was also computed by measuring cosine distances between emotion-specific centroids to quantify the geometric separation among emotion categories.

VGG16 shows a gradually increasing separability trend, peaking at Layer 4. Though the absolute values remain small due to the limited dataset size, the progression implies that emotion-related information accumulates on top of lower-level object features. In contrast, ResNet18 maintains highly entangled emotional information across its early and mid-level layers, with a pronounced increase in separability emerging only in the last layer, indicating that emotional discrimination is deferred to the final decision stage.

RDM analyses reveal that both architectures learn largely object-driven, low-level representations in the early layers, resulting in weak geometric correspondence across models. However, Layers 3 and 4 exhibit strong cross-model similarity, demonstrating that both backbones converge onto a shared mid-level emotional subspace. Despite this shared geometry, ResNet18 subsequently applies a more aggressive non-linear transformation in its final layer to produce a highly separable decision space, whereas VGG16 preserves only a weaker form of separation.

2) Thompson Bandit for Weak Emotion Signal Exploration

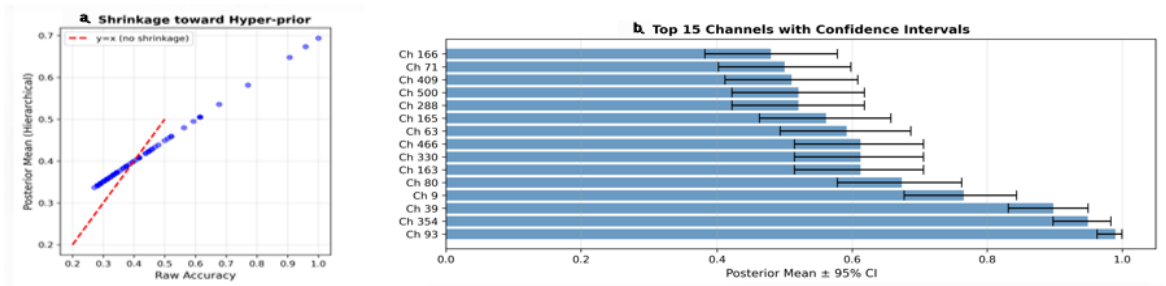
This step investigates how emotion-related information is distributed across individual feature channels within the network, particularly under conditions where strong emotional expressions are not always apparent. In such cases, weak affective cues are prone to entanglement with object-level features. To address this challenge, a Bayesian estimation framework combined with Thompson Sampling was employed to quantify signal reliability and identify statistically meaningful channels that preserve emotion classification performance.

Each channel was evaluated independently using a linear logistic classifier trained on its spatially flattened activations to predict discrete emotion labels. While this setup provides an observable measure of channel performance, it is inherently uncertain due to the limited sample size. To obtain a robust, uncertainty-aware estimate of each channel's true

contribution, a Beta–Binomial formulation was adopted. Observed correct and incorrect predictions were modeled as a Binomial likelihood, and assuming a uniform Beta prior, a posterior distribution was derived for each channel. This posterior captures not only accuracy but also confidence: channels supported by weak evidence retain high uncertainty even when exhibiting moderate accuracy, whereas consistently informative channels yield narrow, confident posteriors. This prevents over-interpretation of noisy signals.

Channel selection was then formulated as a multi-armed bandit problem, where the objective is to maximize emotion discrimination while balancing exploitation of reliable channels and exploration of uncertain ones. Thompson Sampling achieves this by repeatedly drawing synthetic performance samples from each channel's posterior distribution and selecting channels with the highest sampled values. Over iterations, channels with high posterior variance are explored more frequently early on, while stable, high-confidence channels increasingly dominate selection. The resulting convergence behavior reveals how classification performance depends on the number and composition of selected channels.

The Bayesian estimates show that while many channels in the critical layers perform above chance, emotion-relevant information is effectively concentrated in a small subset of channels with high posterior means (often exceeding 0.4). Notably, near-optimal classification performance can be achieved using only a very limited number of such channels, indicating that emotional representations are encoded along sparse, specialized directions in feature space rather than being diffusely distributed. Ablation experiments, in which top-ranked channels were removed, led to marked performance degradation, confirming the functional relevance of these dominant representational axes.



(a) Hierarchical Bayesian inference shrinks noisy channel estimates toward layer-specific priors, stabilizing signal strength while preserving high-performing channels.

(b) A small subset of channels exhibits consistently high posterior means with tight confidence intervals, indicating concentrated emotion-relevant information.

(c) Thompson sampling rapidly converges to the optimal channel subset, confirming efficient identification of informative channels far above chance level.

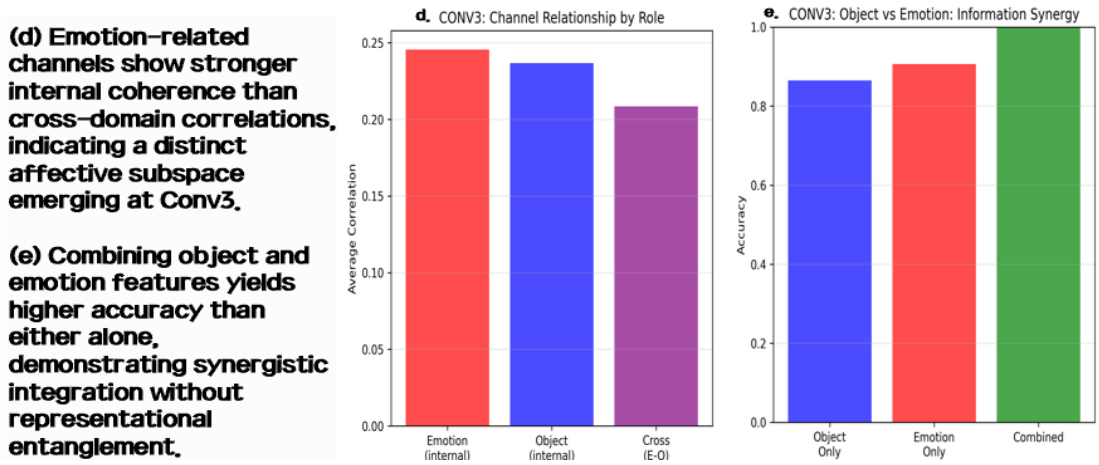
3) Channel Role Discovery and Representational Organization

The purpose of this step is to understand the functional organization through which emotion-relevant information emerges within the hierarchy of the models. Whether the network's internal dynamics genuinely support existence of a structure emotional manifold or the previous representational axes were incidental.

By characterizing the representational geometry across layers with the computation of inter-layer dissimilarity transitions and examining how class centroids separate this becomes an achievable goal. As shown in inter-block transitions VGG16 and ResNet18 both undergo a representational reorganization in previously identified critical layers. As previously discussed, separability enhances slowly within the VGG16 culminating at Layer 4. Whereas ResNet18 waits until the final block to sharply restructure entangled representations. Emotion related structures emerge at transition phases where the model shifts from encoding low-level structures to affective yet abstract dimensions.

Analyzing intra-layer geometry by computing channel-channel dissimilarity matrices, sparsity statistics, and class-conditional activation, a clear difference among channels is evident. High-confidence contributors identified in the previous step form compact activation clusters and exhibit stable responses across samples from the same emotion. While low-importance channels appear in inconsistent subspaces. This emergence suggests that the model looks at specific forms of subspaces with distinct components rather than arbitrary ones.

Despite the revealed difference in the architecture and emotion-specific representational matrices some intermediate layers display high correlation. Alignment between the two models' dissimilarities in Layers 3 and 4 are seen. Indicating both converge on similar geometric representations of emotions after the extraction of low-level perceptual features. This step reveals that emotion representation is both interpretable and systematically organized within CNN hierarchies.

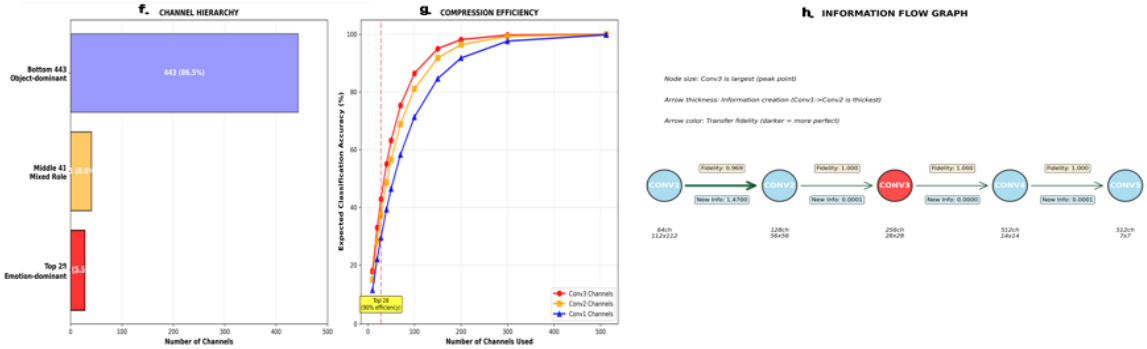


4) Hierarchical Interpretation and Information Flow Reconstruction

The last part reconstructs the hierarchical information flow through which emotional cues propagate across depth. As such, finding where information is manipulated and enlarged to find places functional bottlenecks in the emotion-recognition pipeline.

Using entropy-based stratification and representational compression metrics, the network's channels were divided into hierarchical tiers. The lowest showed low specificity indicating feeble means of emotional content. The following marks reductions in entropy and localize essential regions such as eyes and mouth. It also marks the transitional stage of class-conditional compactness where spatial features are reorganized into higher-level affective cues. The highest follows suit in reducing representational variance. These channels display emotion-selective activation patterns and give pivotal decision-making cues.

Cross-layer interaction is evaluated through activation covariance and gradient-based dependency analysis. Results reveal an information flow graph of directed structure where high-level channels act as essential parts in decision process led up by mid-level channels. GRAD-CAM visualizes important regions associated with emotional diagnosis. Which are in consensus with SHAP attributions. Making point how pivotal positions in the system functionally have a role in changing perceptual input into affective judgment. As the emotion-relevant information gets compressed across different layers concentrates into a compact low-dimensional bottleneck. Meaning once relevant information has been compressed into a handful of meaningful dimensions, architectural differences no longer play a significant role in the geometry of representation. These findings integrate representational geometry, hierarchical compression, and attribution analysis into a unified interpretive account of how CNNs encode and utilize subtle emotional signals.



(f) Most channels remain object-dominant, while a small subset of mid-level channels becomes emotion-selective, indicating sparse and hierarchical affective encoding.

(g) Emotion classification accuracy saturates rapidly, with only 28 Conv3 channels capturing over 90% of the full-model performance.

(h) Information flow analysis reveals Conv3 as a bottleneck layer where emotional information peaks before redistribution in deeper layers.

3. 연구의 의미

This study demonstrates that emotion-relevant information is not uniformly distributed across visual hierarchies but emerges progressively in architecture-dependent ways. Across both VGG16 and ResNet18, emotional separability increases from intermediate layers, reflecting core principles of biological vision in which representations evolve from low-level sensory features to higher-order affective abstractions.

Comparative analysis reveals that architectural bias shapes emotional encoding: VGG16 exhibits gradual accumulation of affective separability, whereas ResNet18 compresses emotion-relevant signals into later stages. This contrast parallels differences between feedforward integration and recursive processing observed in biological visual–emotional pathways.

By incorporating Bayesian inference and Thompson sampling, the study further shows that emotional perception relies on a small subset of highly reliable channels, supporting a probabilistic view of affective encoding consistent with predictive processing theories in neuroscience. Inter- and intra-channel dependency analyses reveal directed information flow, suggesting that low-level visual features provide a structural scaffold upon which affective meaning is consolidated in deeper representations.

Overall, these findings support the view that emotion is not a static label but a transformation of perceptual statistics into structured affective manifolds. By unifying representational geometry, uncertainty modeling, and hierarchical dynamics, this work bridges computational representation learning with biological theories of emotion as an emergent, self-referential process.

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